A Stacking Approach for Arctic Sea Ice Lead Classification (SALC) via Sentinel-1 SAR Imagery

Xi Zhao, Yi Zhang, Jiaxing Gong, Yifan Wu, Xiao Cheng, and Georg Heygster, *Member, IEEE*

***Abstract*— Leads are linear fractures formed by the deformation of the ice cover and are crucial areas for Arctic heat exchange between the ocean and the atmosphere. Accurately identifying leads is essential for studying climate change, polar ecosystems, and shipping navigation. Recent research on leads detection methods developed using synthetic aperture radar (SAR) tend to ignore the developmental stage of the leads and the distribution context of leads [first-year ice (FYI) or multiyear ice (MYI)], which hinders the generalizability of these methods. In this article, we propose a stacking approach for lead classification (SALC) integrating five specialized learners for five typical lead conditions. Ablation experiments reveal the effectiveness of each component of SALC: 1) the optimal preprocessing method is to input the incidence angle as a feature and apply the advanced thermal noise removal. In addition, the effect of inputting the incidence angle is significantly stronger than the incidence angle correction for original images; 2) the SALC performs best when the data features include polarization, texture, and upscaling features; 3) XGBoost is the most suitable meta classifier as compared to support vector machine (SVM) and random forest (RF). The results show that SALC outperforms both single learners and traditional classifiers across various conditions and quality criteria. In addition, SALC can tolerate at least 15% of erroneous samples.**

***Index Terms*— Arctic Ocean, sea ice leads, Sentinel-1, stacking algorithm, synthetic aperture radar (SAR).**

I. Introduction

LEADS in Arctic sea ice are linear cracks within the ice cover, ranging from below 1 m to several kilometers in width [1]. They are usually formed as a result of ocean currents, winds, and other forces causing ice fracturing and deformation, while in the frigid Arctic winter, they freeze over rapidly. According to the World Meteorological Organization (WMO), leads encompass both open water and thin ice layers less than 30 cm thick. Although open leads cover only 1%–2% of the central Arctic region in winter, they contribute to more than 70% of the sea-air heat exchange [2], [3]. The heat flux over open leads is two orders of magnitude greater than over ice cover [4]; as leads freeze (<1 m thickness), the exchange weakens but still remains several times higher than over multiyear ice (MYI) [3], [4]; thus, leads play a significant role in energy transfer in the Arctic. In recent years, due to Arctic amplification, sea ice in the region has become more fragile and mobile, causing a greater shift from MYI to seasonal ice, with cracks forming leads more frequently [5], [6], [7]. This implies that the role of ice cover in heat exchange within the Arctic Ocean is becoming increasingly important and that the springtime extent of leads might even serve as an indicator of the summer sea ice range [8]. The increase in leads may also impact biological activity [9], as Arctic wildlife—such as polar bears, seals, and walruses—often depends upon leads. Last, leads are crucial for polar navigation [10], as their increase presents new potential for Arctic shipping routes.

Since the last century, researchers have explored the development process and electromagnetic characteristics of Arctic leads by means of field observations and laboratory simulations [11], [12]. Under cold conditions, the ice within leads grows rapidly and exhibits high salinity, with surface features primarily dominated by frost flowers [11]. Laboratory simulations have shown that ice slush forms on the surface of lead ice, providing a growth base for frost flowers. Even if the more fragile frost flowers are manually removed, the remaining snow slush layer still highly influences the backscatter of C-band radar [12].

Field surveys in the Arctic are extremely costly, which makes satellite remote sensing an important tool for sea ice observation. In recent years, remote sensing research on Arctic leads has primarily relied on thermal infrared and microwave imagery. By exploiting the differences in albedo and brightness temperature between leads and sea ice, leads can be extracted using methods such as thresholding [13], [14]. Several lead products derived from thermal infrared [15], [16] and microwave images [17], [18] are now available. These products, however, typically have a low spatial resolution (≥1 km) and are based on daily composites, which results in many leads being undetected and the width of detected leads being overestimated. Additionally, thermal infrared imagery is affected by cloud cover [19], inevitably leading to gaps in lead monitoring in certain regions.

Synthetic aperture radar (SAR) can provide all-weather, day-and-night imaging, allowing for large-area, high-resolution monitoring of ice leads. SAR imagery, however,

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is prone to speckle noise due to its coherent imaging process, making it challenging to segment images using thresholding methods [20]. Murashkin et al. [21] and Zakhvatkina et al. [22] applied machine-learning methods, and specifically support vector machines (SVMs) and random forest (RF), to identify leads from C-band dual-polarization SAR images using texture features. Besides, the Gray Level Co-occurrence Matrix (GLCM) has been recognized as an effective method for SAR image classification [21], [22]. Recent research has focused on deep-learning models. For instance, Murashkin and Frost [23] improved the U-Net++ network architecture to identify open-water leads and wind-driven rough leads from Sentinel-1 imagery. Zakhvatkina et al. [24] used U-Net to process Sentinel-1 images, achieving more robust results compared to the RF. Liang et al. [25] proposed an entropy-weighted network (EW-Net), a modified version of U-Net, for lead detection under low-wind conditions. Huang et al. [26] employed a dual-branch encoder U-Net model fusing polarization features and GLCM information to identify both sea ice types and leads. Chen et al. [27] introduced the SILDET method for detecting leads across different freezing stages in the entire Arctic. Qu et al. [28] compared multiple RF models based on polarization and texture features, identifying the most sensitive model for detecting narrow leads in Sentinel-1 imagery.

Previous studies have focused on specific regions, with limited consideration of the developmental stages of leads. This makes it difficult to ensure the generalizability of the method and its ability to map leads across the entire Arctic. For example, Murashkin and Frost [23] and Zakhvatkina et al. [24] considered open-water leads and did not account for the freezing state of the leads. Liang et al. [25] had designed for open-water leads under low-wind conditions. In the cold season, leads freeze rapidly, and thus, their models can only detect a small proportion of open-water leads. The training dataset in [26] was limited to the Beaufort Sea, and its results showed that many leads were misclassified as first-year ice (FYI), with leads occupying only a small portion of the identified areas. Qu et al. [28] conducted in MYI regions and did not consider frozen leads. Arctic sea ice can be roughly divided into FYI and MYI. FYI is the sea ice that melts completely during the first summer, and it exhibits distinct features in SAR imagery compared to MYI [20]. \Training samples should, therefore, take into account the different contexts, irrespective of using traditional machine-learning algorithms or deep-learning approaches.

In this article, we aim to propose an effective method for detecting Arctic leads that are surrounded by different type of sea ice and at different developmental stages. We apply the concept of stacked generalization [29], where multiple base learners are constructed, and their results are fed into a meta classifier to obtain the final output. Similar methods were effective in previous researches [30]. In this stacking approach for lead classification (SALC), each base learner is specialized on a particular condition, making the entire stacked model function like a team of experts, thereby increasing the reliability of the final results. The proposed method uses pixels as the smallest training unit, considering both the backscatter features of individual pixels and the spatial information of textures, enabling the identification of both narrow and wide leads to the greatest extent.

II. Data

A. Sentinel-1 Image

Sentinel-1 is a dual-polarization C-band SAR satellite mission developed and operated by the European Space Agency (ESA). Its primary objective is to provide all-weather radar imaging of the Earth's surface to detect surface changes. The Sentinel-1 constellation comprises three satellites, namely Sentinel-1A, Sentinel-1B, and Sentinel-1C, launched in 2014, 2016, and 2024, respectively. Unfortunately, Sentinel-1B ceased its mission in 2022 due to a malfunction. It was not until the launch of Sentinel-1C in December 2024 that the constellation regained its dual-satellite configuration. The dual-satellite system increases the frequency of Earth coverage, with a revisit period of six days, which can be reduced to two days in the Arctic region, covering up to 87.5°N latitude. Although the coverage frequency of Sentinel-1 over polar regions declined after the death of Sentinel-1B in 2022, Sentinel-1C has now begun providing data, gradually restoring high-frequency observations of the polar areas. Unlike optical imagery, Sentinel-1 is not affected by polar night or cloudy weather, making it ideal for capturing changes in ice leads in the Arctic's variable weather conditions. Sentinel-1 offers four imaging modes: Interferometric Wide (IW), Strip Map (SM), Extra Wide (EW), and Wave (WV). In this study, we use the Ground Range Detected (GRD) product under the EW Swath mode. This product converts radar slant range to ground range, losing phase information in the process. The EW GRD images have a swath width of 400 km and a pixel spacing of 40 m, with an incidence angle range from 18.9° to 47°. In polar regions, EW images are available in HH and HV polarizations. This study is based on images in these two polarizations, with detailed processing methods described in Section III-A. The information on the Sentinel-1 images used for training and independent validation is detailed in Table I.

B. Sentinel-2 Image

Sentinel-2 is an optical imaging satellite series under the ESA's Copernicus program, consisting of a twin-satellite configuration (A/B), with a revisit period of ten days. The 180° phase difference between the satellites reduces the revisit period to five days. The primary mission of Sentinel-2 is to conduct high-resolution Earth observation using a multispectral imaging sensor [Multispectral Instrument (MSI)]. The MSI sensor on Sentinel-2 covers 13 bands, ranging from visible light to near-infrared and shortwave infrared (SWIR), with wavelengths from 443 to 2190 nm. In this study, we used the near-infrared Band 8 from Level-2 data, with a spatial resolution of 10 m, to match and compare with Sentinel-1 imagery. This band was used to enhance the understanding of lead conditions and to validate the classification results. The Sentinel-2 images corresponding to independent validation Sentinel-1 images are detailed in Table I.

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TABLE I
DETAILED INFORMATION OF THE IMAGE DATA USED

No.	Acquisition time	Acquisition time of compared Sentinel-2 images (Black boxed in Fig. 1)	Location
Independent validation images			
C1A	Feb. 27, 2019 15:28	Feb. 28, 2019 21:04	Beaufort Sea
C1B	Mar. 28, 2019 02:33	Mar. 28, 2019 08:06	Kara Sea
C2A	Apr. 22, 2019 16:16	Apr. 22, 2019 22:00	Beaufort Sea
C2B	Mar. 29, 2019 22:27	Mar. 29, 2019 04:15	Laptev Sea
C3	Apr. 22, 2019 18:38	Apr. 22, 2019 20:18	Lincoin Sea
C4	Mar. 26, 2021 09:22	Mar. 26, 2021 16:38	Wandel Sea
C5	Mar. 25, 2019 17:21	None	Beaufort Sea
Images for training			
①	Feb. 27, 2019 15:28	/	Beaufort Sea
②	Mar. 09 2021 12:28	/	Central Arctic
③	Jan. 20 2020 15:16	/	Central Arctic
④	Dec. 22 2020 18:55	/	Lincoin Sea
⑤	Mar. 25 2019 15:59	/	Beaufort Sea

C. Other Data

This study used sea ice age data [31] from the National Snow and Ice Data Center (NSIDC). This dataset is stored in NC files with a projection of EPSG:3408 and a spatial resolution of 12.5 km. This dataset has been updated weekly until the end of 2023. This dataset uses values from 0 to 16 to represent the sea ice age, where 0 denotes open water, 1 represents FYI, and the remaining numbers indicate the age of the MYI. We calculated the frequency of MYI occurrence across the entire Arctic from January to April during 2014–2023. During this period, the frequency of MYI occurrence in the Arctic winter since the launch of Sentinel-1 can, thus, be analyzed. This was overlaid on the sea ice extent data, also derived from this dataset, to indicate the distribution of sea ice across the Arctic, which served as the basis for classifying lead conditions (see Fig. 1).

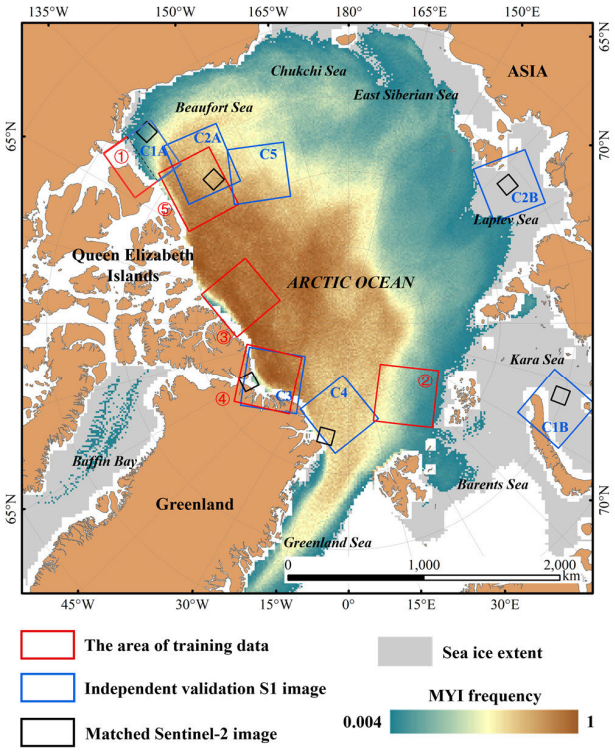


Fig. 1. Distribution of the imagery used in this study overlaid on the frequency of MYI occurrence. The gray area represents the sea ice extent during this period; the red boxes ① –⑤ are the area used for training the model in five conditions; C1–C5 (blue boxes) are the independent validation Sentinel-1 images covering the five conditions; and black boxes are the corresponding Sentinel-2 images matched with the independent Sentinel-1 images.

III. METHODOLOGY

In this section, we will provide a detailed description of the preprocessing of Sentinel-1 imagery, label generation, model structure, and model evaluation. The data processing and method validation workflow is shown in Fig. 2.

A. Sentinel-1 Image Preprocessing

The preprocessing of Sentinel-1 imagery included thermal noise removal, radiometric calibration, speckle noise removal, and decibelization. Although the ESA provided a graphical processing tool (SNAP) for Sentinel-1 data, we did not directly use it for preprocessing because the noise description files provided by the ESA did not fully address the impact of the scalloping effect and thermal noise (in this article collectively called "thermal noise") on HV imagery [32], [33], [34]. As shown in Fig. 3(a), the scalloping effect is a strip-like noise distributed regularly in the azimuth direction. In this study, all HV polarization images were denoised using the method proposed by Sun and Li [34]. This method improved upon the approach by Park et al. [32], dividing each swath of the image into smaller subblocks and designing a variance factor to filter out uniform blocks for the calculation of scaling and balancing factors. The method adaptively reconstructs a 2-D noise field for each image. Subsequently, scaling parameters for each swath were calculated based on the subblocks to remove the residual multiplicative noise along the swath boundaries. During the noise removal process, the images have already

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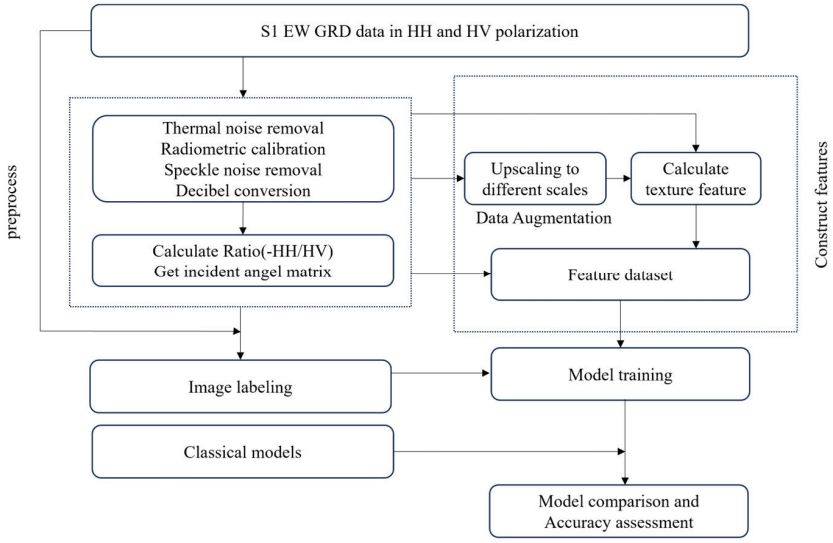


Fig. 2. Data processing and lead detection method flowchart.

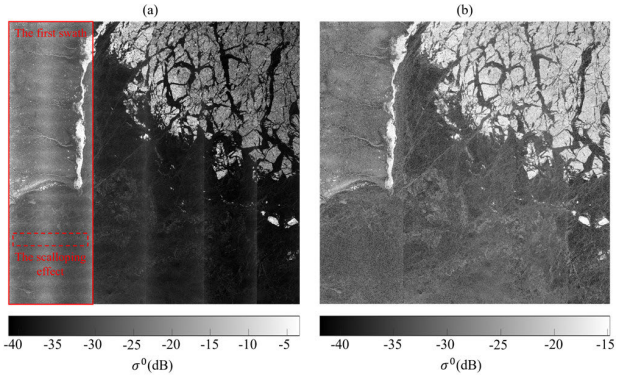


Fig. 3. HV polarization of image C1A denoising (display with 2% linear stretch). (a) Original HV polarization image, heavily affected by thermal noise. (b) HV polarization image after applying the improved thermal noise removal method. Image ID: S1A_EW_GRDM_1SDH_20190227T152810_20190227T152910_026117_02E9F0_3D16.

undergone calibration, as outlined below

$$\sigma^0_{\mathrm{SN}} = \frac{\mathrm{DN}^2 - \mathrm{no}}{A^2} \tag{1}$$

where, DN represents the original digital quantization value of the image, no is the reconstructed 2-D noise field, A is the calibration parameter provided by ESA, and σ^0_{SN} is the backscatter calibrated to sigma0 after noise removal. During denoising, the reconstructed noise field was also applied to the HH polarization images. Fig. 3 shows the effect of denoising on the HV polarization of image C1A. In the HV image, the backscatter of FYI is significantly lower than that of MYI, which results in a greater impact of thermal noise on FYI in the low signal-to-noise ratio of the HV polarization. The noise is most severe in the first swath of the HV image, where the backscatter value of FYI is even influenced to the point of resembling that of MYI. After denoising, the impact of noise is greatly reduced. The backscatter of FYI in the first swath is restored to its normal level. Because of the severe thermal noise in the first swath, a small portion of the noise, however, remains challenging to eliminate.

During SAR sensing, the echo signals of adjacent pixels undergo random variations due to coherence, leading to speckle noise in SAR imagery [35]. Specifically, regions with more severe speckle contamination appear brighter in the image, which can affect target recognition; therefore, we applied Lee filtering [36] to the denoised images with a window size of 7 × 7. Finally, to enhance the contrast, we applied decibelization to the speckle-removed images.

The backscatter of HH polarization in SAR imagery is highly sensitive to the incidence angle. In previous related studies, incidence angle correction was commonly applied, such as in [27] and [28]. Both studies obtained empirical correction coefficients for the incidence angle using statistical methods. Qu et al. [28] performed a global correction for the incidence angle dependence across the entire region, while [27] separately analyzed the incidence angle dependence for sea ice and water and determined which correction coefficient to apply for each pixel using a threshold. Some studies [37], [38], [39] have also pointed out that different ice types, however, exhibit varying degrees of dependence on the incidence angle. Given the diversity of ice types in the Arctic and the different stages of lead development, it is difficult to accurately perform incidence angle correction for each pixel; thus, we did not directly perform incidence angle correction on the HH polarization in this study. Instead, we extracted the incidence angle matrix from the annotation file and included it as a feature input into SALC.

B. Constructure of Label

The spatial distribution of sea ice type in the Arctic is highly regionalized. The region north of the Queen Elizabeth Islands and Greenland in Canada is a MYI area. The MYI in this region drifts to the Beaufort Sea and the Chukchi Sea due to the Beaufort Gyre, with some parts exiting through the Bering Strait while others return to this area. As a result, the Beaufort Sea and Chukchi Sea are typical mixed areas of FYI and MYI. As shown in Fig. 1, we used the ice age data released by NSIDC to calculate the frequency of MYI occurrence from

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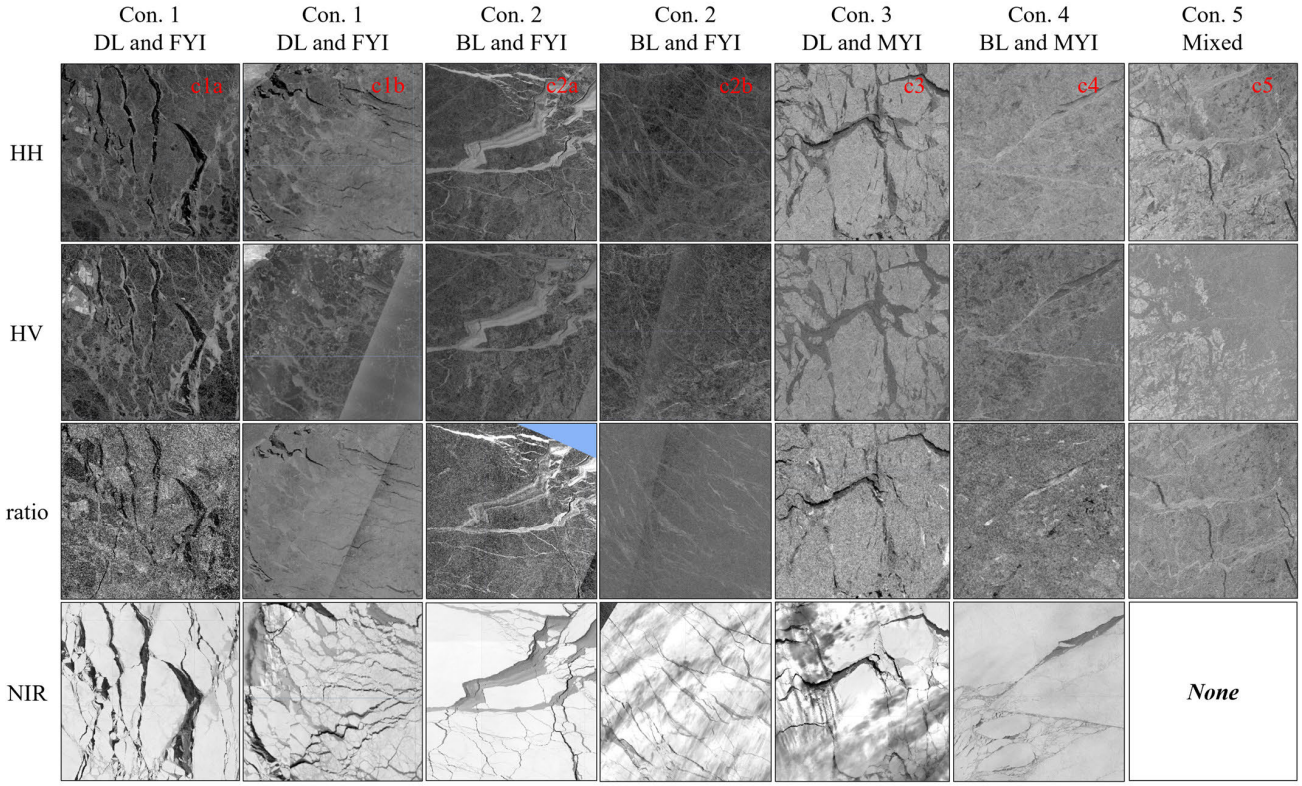


Fig. 4. Sentinel-1 and Sentinel-2 image pairs of the five conditions, illustrating the characteristics of each condition in SAR. The HH and HV polarizations are from the original Sentinel-1 images, the ratio is calculated using (−HH/HV), and the NIR images are from Sentinel-2 Band 8. c1–c5 are subimages of C1–C5, respectively. NIR images are from compared black boxes in Fig. 1.

January to April during 2014–2023 overlaid on the maximum ice extent during this period. MYI is consistently present in the northern coast of Canada and the central Arctic Ocean, while the ice cover is predominantly FYI, with little intrusion of MYI in the Laptev Sea, Kara Sea, Barents Sea, and Baffin Bay. In the eastern Arctic Ocean, the frequency of MYI occurrence is mostly below 20%, as shown by the cyan-green area in the figure which is considered to have occasional MYI presence. The frequency of MYI in the Greenland Sea and Fram Strait is similar to that in the Beaufort Sea, but the distribution of both types of sea ice in the Beaufort Sea has a clear boundary, where FYI in the Greenland Sea and Fram Strait is more randomly distributed among the MYI.

We classify leads into bright leads (BL) and dark leads (DL), which appear brighter or darker than the background, respectively; therefore, based on the characteristics of the leads and the distribution of sea ice background, we set five conditions illustrated by seven examples in Fig. 4.

1) *Con. 1:* FYI area with more DL.
2) *Con. 2:* FYI area primarily dominated by BL.
3) *Con. 3:* MYI area with more DL.
4) *Con. 4:* MYI area primarily dominated by BL.
5) *Con. 5:* FYI and MYI mixed areas with both bright and DL.

Both Con. 1 (c1a and c1b) and Con. 3 (c3) show DL. DL are newly formed leads with open water or very thin ice layers inside. This thin ice layer has a smooth surface, so the backscatter of DL is very low in both HH and HV polarizations. In Fig. 4, the DL in subimages c1a and c3 represent open water and newly frozen thin ice, respectively. In c3, the lead has begun to brighten, and its freezing state can be observed from the NIR image. The ratio, as a derived backscatter feature, does not have a clear physical meaning but enhances the interclass differences. From c1a, c1b, and c3, it can be seen that DL shows a low value of the ratio below the surrounding background.

Both Con. 2 (c2a and c2b) and Con. 4 (c4) show BL. BL refers to the frozen leads. The open water exposed in the lead freezes and thickens, with salt being precipitated out to form frost flowers, which increases the backscatter in both HH and HV polarizations [39], [40], [41]. As freezing continues, the backscatter of the thin ice in the lead decreases again because the frost flower decays, the ice layer thickens, and the surface becomes smoother, eventually transforming into FYI. Since BL may have gone through different freezing states, it is common to see a layering pattern parallel to the edge of the lead, as shown in c2a. As shown in c2b, when a BL freezes and thickens, its brightness on SAR darkens, and its boundaries may become blurred in both HH and HV. In this case, the ratio is needed to accurately capture its boundaries. Regardless of how a BL changes, it will always be brighter than the surrounding background in the ratio until it becomes FYI.

Con. 5 in c5 represents the lead conditions in the mixed ice zone. We can observe many linear features in the HH, HV, and ratio images. FYI undergoes a transition process where a BL turns into FYI, and it is difficult to determine this transition point from the imagery. After some BLs transition into FYI,

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their surface frost flowers disappear and may become smooth. The dB values in HH, HV, and the ratio are all lower than the surrounding background, making it easy to confuse with DL.

Certainly, the actual development of leads can involve more complex deformations. For instance, during the freezing process of DL, the rapid drift of sea ice may cause the thin ice within the leads to deform, resulting in phenomena such as finger rafting and small ridges. These deformations can enhance the backscatter of the leads. Additionally, snow accumulation on the leads can form wind slabs and ice lenses, increasing volume scattering and making the leads appear brighter in imagery. Given the dynamic and complex nature of the Arctic environment, these factors may also contribute to the leads being characterized as BL.

We separate the training and validation images independently. The model was trained by samples derived from areas bounded by red boxes in Fig. 1. The classification results of validation subimages c1a, c2a, c3, c4, and c5 (from images C1–C5 in Fig. 1) are compared with corresponding Sentinel-2 images. As for the training data, we first made region labels and then took pixel samples within the region labels. So, we have a large number of pixels for training and testing (7:3 ratio shown in Table II). The dataset generation was done using Pandas package of python, in which data was shuffled before divided, ensuring the randomness of the data.

C. Stacking Approach

In the labeled samples, there is a significant overlap in backscatter values between different ice covers, posing a great challenge for accurately distinguishing leads from the background. It is difficult to ensure that a single model will perform well across all conditions. This is illustrated by the "no free lunch" theorem proposed by Wolpert and Macready [42]: "If an algorithm performs well on a certain class of problems, then it necessarily pays for that with degraded performance on the set of all remaining problems." For example, if a model designed for detecting BL in a MYI area performs poorly on detecting DL in an FYI area, we can more simply understand this as the model not having learned enough to handle other conditions.

In order to address this issue, we adopted the idea of stacking generalization. Based on the characteristics and distribution of lead development, we divided the conditions into five categories, Con.1–Con. 5 (see Fig. 4). Corresponding training samples were extracted from each condition to generate specialized learners, and the outputs of these learners were then combined to form the final output. The proposed model, called "SALC," has a two-layer structure: a base learner layer and a meta-classifier layer. Based on experience, base learners in stacking models need to be strong models capable of learning the deep relationships between input data and labels, while the meta classifier should be a model with good interpretability, as it integrates the outputs of the base learners and prevents overfitting. We, therefore, chose multilayer perceptron (MLP) as the base learner and XGBoost as the meta classifier.

The structure of the MLP in the base learner is shown in Fig. 5. The MLP consists of an input layer, hidden layers, and an output layer. It is a feedforward neural network that mimics the structure of biological neurons, relying on the powerful nonlinear learning capability of multiple hidden layers to model the relationship between input and output. Every two neurons in successive layers are fully connected through logistic regression; thus, it is also referred to as a fully connected network. Our MLP model consists of five hidden layers, with 512, 256, 128, 64, and 32 neurons, respectively, following the propagation direction. Batch normalization layers were added after the input layer and each hidden layer to accelerate the training process. The activation function for both the input and hidden layers is "rectified linear unit" (ReLU), which introduces nonlinearity to allow the network to learn more complex relationships. The output layer has no activation function. The optimizer used is Adam [43], which adapts the learning rate and converges quickly, making it well-suited for large datasets like the case in this study. The loss function used for training is weighted cross-entropy. Since the sample distribution of different ice cover classes is imbalanced across various conditions, we assign weights based on the proportion of each ice cover class in the samples, to best reflect the model's performance. The weighted cross-entropy function is

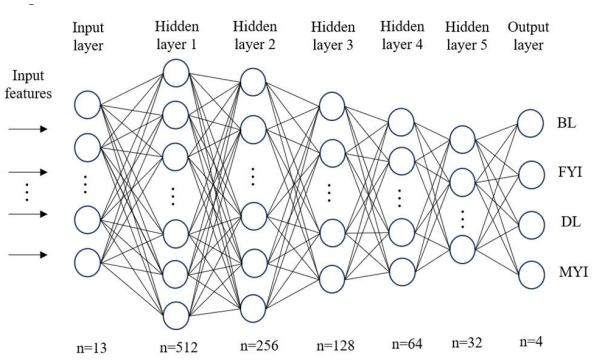


Fig. 5. Structure of the base learner MLP.

$$\text{Loss}(p, y) = -\sum_{i=1}^{C} \frac{N_i}{N} \cdot y_i \cdot \log p_i \tag{2}$$

where C is the total number of classes, N is the total number of samples, N_i is the number of samples in the i class, y_i is the true label of the i class, and p_i is the predicted probability of the i class. The model is built using PyTorch [44], and each model is trained for 100 epochs, which means that the training dataset is iterated through 100 times during the model fitting.

XGBoost [45] is a Boosting method based on decision trees derived from the Gradient Boosting Decision Tree (GBDT). Similar to the classic RF model, XGBoost combines multiple weak learners to form a strong learner, where multiple trees collaborate to make decisions. The key difference is that in RF, the final result is derived by averaging or voting among the decision trees, and the trees are independent of one another. In contrast, XGBoost is a forward additive model where each tree estimates the residual of the previous tree, meaning that the trees are interconnected. Additionally, XGBoost incorporates several unique improvements, such as regularization and feature-level parallelism, to enhance generalization and facilitate rapid convergence.

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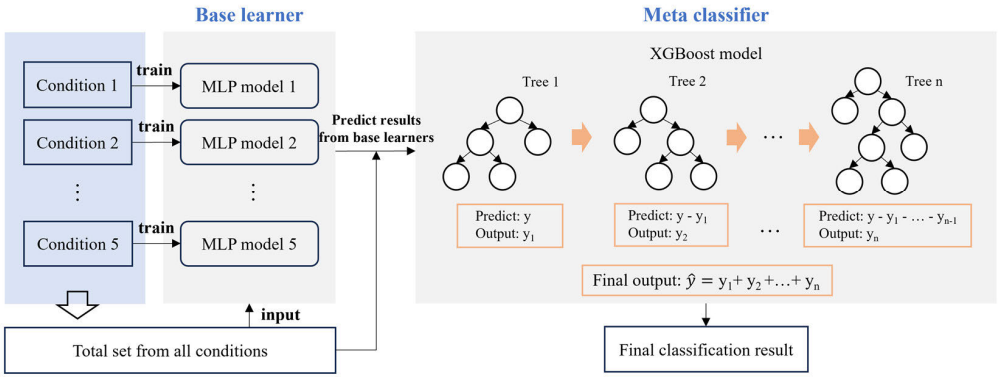


Fig. 6. Structure of SALC. The number n of decision trees in the meta classifier is 100.

After determining the structure of the SALC, we constructed the input features, including polarization, texture, and upscaling features. We employed GLCM as an additional feature to improve the classification accuracy of leads. GLCM has been shown to be effective for land cover classification in many studies, as it quantifies the relationship between a central pixel and its surrounding pixels to describe the texture characteristics of an image. Several parameters need to be set when calculating texture features using GLCM: window size, step size, angle, and gray level [46]. Since the orientation of leads can vary in different directions, we averaged the results from four angles: 0°, 45°, 90°, and 135°. The other parameter settings are window size 7 × 7, step size 1, and gray levels 16. We extracted eight texture features, including Mean, Variance, Homogeneity, Contrast, Dissimilarity, Entropy, Second Moment, and Correlation. Because we already had images from the HH, HV, and ratio bands, calculating texture features separately for each would result in data redundancy. We, therefore, performed a principal component analysis (PCA) [47] on the three bands and used the first principal component for texture calculations. The final feature dataset includes HH polarization, HV polarization, ratio, the eight texture features, the first principal component of the three polarization bands, and the incidence angle matrix. In addition, we employed a bilinear interpolation resampling method to upscale the original image pixels at the resolution of 40–100 m, 250 m, and 500 m and generated new samples from the upscaled images as data augmentation, supplementing the original samples. In these upscaled images, the variance of samples within each class will decrease, which helps the model to ignore certain noise.

In this study, training sets of Con. 1–Con. 5 (red boxes in Fig. 1) were used to train MLP model 1–5 as base learners (see Fig. 5). Then the training sets from all conditions as a whole were entered into the five base learners. In this way, for each pixel sample, we can obtain five classification results from five different base learners. These five results, along with the original features such as backscatter and texture, were put into XGBoost to obtain the final result (see Fig. 6). Instead of directly inputting the predicted class code (BL: 1; FYI: 2; DL: 3; MYI: 4), we input the product of the class probabilities and their corresponding code to increase the amount of information.

D. Evaluation of SALC

In order to assess the performance of the SALC, both qualitative and quantitative evaluations are conducted. We first applied SALC to independent SAR images C1A, C2A, C3, and C4 for different conditions. The detailed visual comparison was illustrated by the subimages c1a, c2a, c3, and c4, respectively. Next, we applied SALC and the single classifiers to C5 image for the most complex condition 5. Finally, we compared their results and verified the advantages of SALC.

In the quantitative evaluation, we chose commonly used metrics based on the confusion matrix [48], including precision, recall, overall accuracy, F1-score, and the Kappa coefficient. In the confusion matrix, the predicted outcomes are typically represented by true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Recall indicates the model's ability to identify all positive cases. A high recall means the model can identify as many positive samples as possible, but it may lead to false positives. Precision, on the other hand, reflects the reliability of the model's positive predictions. A high precision means fewer false positives among the samples identified as positive. Sometimes, precision and recall can be in conflict, and the F1-score is a combined measure that considers both metrics. Overall accuracy (OA) refers to the proportion of correctly classified samples out of all samples. The Kappa coefficient is a statistical measure used to assess consistency, with values ranging from $[-1, 1]$. A value closer to 1 indicates a higher consistency between the predicted and actual results.

Additionally, for a deeper analysis of SALC, an ablation (omitting processing steps) experiment was conducted to investigate the impact of each component of SALC.

IV. Results

A. Comparison With Classification Results of Sentinel-1 and Sentinel-2

Fig. 7 shows four SAR images of the four base conditions and the corresponding NIR images (the corresponding HV and ratio images are shown in Fig. 4), along with the classification results using SALC. The subimage c1a is taken from the FYI area within the Beaufort Sea mixed ice zone. Most of the sea ice in this region is fragile and prone to deformation, with cracking and refreezing phenomena being common. As a result, the classification shows some DL and BL patches.

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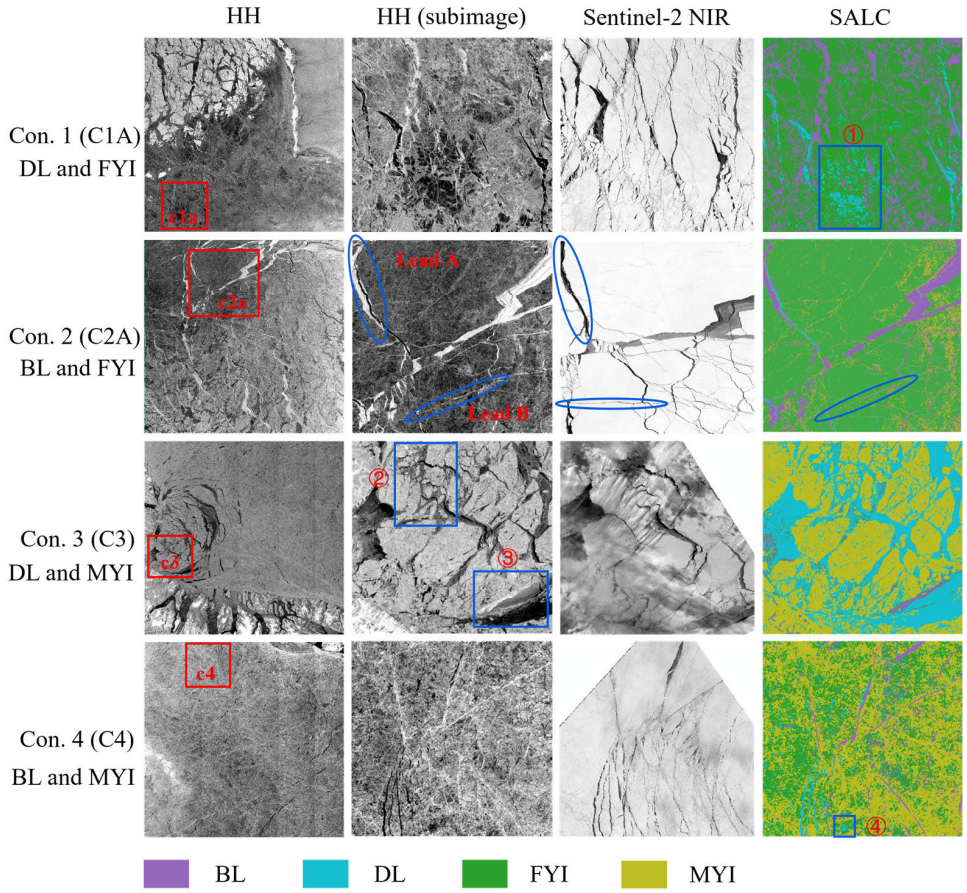


Fig. 7. Classification results of the four base conditions compared with the Sentinel-1 and Sentinel-2 images. The first column HH shows the original SAR images of C1–C4; the second column displays the c1–c4 subimages corresponding to the red boxes in the HH column; the Sentinel-2 NIR column shows the NIR band of the matched Sentinel-2 images; and SALC column presents the classification results of SALC.

Based on the interpretation of SAR images, the presence of these areas is reasonable. The target lead class in this region is DL, and SALC detected nearly all the DL visible in the SAR images (visual inspection). And the results are highly consistent with the matching Sentinel-2 imagery. The classification result in region ① is mainly DL (blue rectangle), but according to NIR imagery and visual interpretation, this area should actually be FYI; that is because the backscatter and texture in this region closely resemble the DL in the SAR image.

Although the time difference between the c2a image pairs is less than 6 h, the freezing stages of the leads may not be completely consistent. Lead A (within the blue ellipse) is clearly visible as DL in the NIR image, but in the SAR image, most of it has frozen into BL. This subimage also originates from the Beaufort Sea, but it shows fewer discrete ice cover types in the classification results. There are still instances, such as Lead B, where small leads are, however, identified as discontinuous and fragmented patches. Additionally, some leads are surrounded by MYI in the classification results, which could be due to the blurred boundaries of the small leads. Overall, SALC is able to accurately identify most of the leads in c2a.

Subimage c3 is located north of Greenland in the MYI area, making the ice solid and firm. The spiraling leads in the figure are typical of DL on MYI and are likely influenced by ocean dynamics, especially tides, often showing spiral cracking patterns. The matching optical image was captured within 3 h after the SAR image, and it shows that some of the leads have completely frozen. In region ② (see Fig. 7), most of the DL that started freezing are no longer visible in the optical image. The BL in region ③ has already completely frozen, and DL has begun to freeze as well. Because of the rapid freezing of leads in this area, there are different stages of frozen leads. The model performs well in this condition, successfully identifying almost all of the leads.

Subimage c4 is from the northward of the Fram Strait, partly also within the MYI area. The sea ice in this region is similarly solid, and the leads formed by cracking are mostly narrow and elongated. The image was captured in March after the concentration of MYI in this area had been decreasing since December of the previous year. As a result, the background of the subimage contains a significant amount of FYI. The classification results for c4 accurately identified most of the leads visible in the SAR image, but there is still a small area of FYI mistakenly identified as DL [e.g., Region ④ (see Fig. 7)].

B. Comparison With Single Classification Models

Subimage c5 is located in the Beaufort Sea, where the sea ice is fragile and diverse in stages, making it the most complex among all conditions. We first applied the base learners (MLP)

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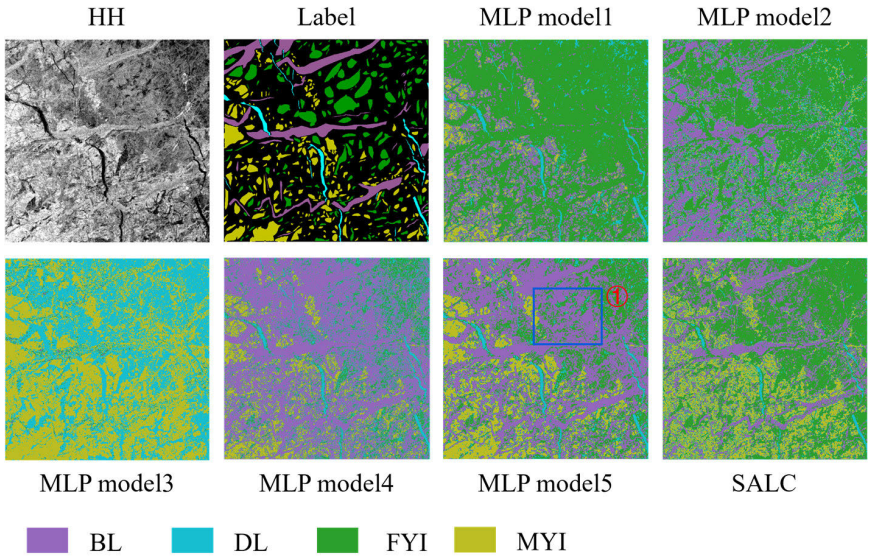


Fig. 8. Classification results of SALC and the specialized learners for each condition applied to the subimage c5.

trained on each condition to c5 and compared them with SALC.

Fig. 8 shows subimage c5, the manual labels, and the classification results of the specialized learners from each condition applied to this subimage. The same classes in the manual labels and model classification results are represented by similar colors: purple for BL, blue for DL, green for FYI, and yellow for MYI. MLP models 1–5 are the specialized learners for Con. 1–Con. 5, respectively. Because each learner was trained on data from a single condition, they exhibit different overfitting issues when applied to the more complex Con. 5. MLP model 1 is used for identifying DL in the FYI region, and it is more sensitive to FYI and DL. In the results, it accurately identifies the DL in c5, but overestimates the FYI and fails to accurately identify MYI and BL. MLP model 2, on the other hand, sacrifices the recognition of DL and MYI in order to enhance the identification of FYI and BL, leading to an overestimation of FYI and BL. Similarly, MLP model 3 and MLP model 4 do not perform well when they face conditions outside their training data. MLP model 5 is the specialized learner designed for these complex conditions, and it should ideally perform the best in c5; however, it still shows some degree of misclassification. For example, in region ① (see Fig. 8), it should be identified as FYI based on visual interpretation but is classified as a BL. This may be due to the similarity of certain FYI features to BL in this condition, which misleads the learner. SALC improves the performance on this part, although it does not completely solve the issue. SALC integrates the experiences of multiple conditions to handle these ambiguous regions, similar to consulting multiple experts. As a result, it is more reliable in classifying these complex regions than single learners.

We then compared SALC with three classical models: SVM, RF, and MLP. The training dataset and input features for the three traditional models are the same as SALC, using the combined dataset from all conditions. The MLP structure is the same as that of the base learner. Traditional SVM, however, struggles to handle such large amounts of data, so we simulated a nonlinear SVM using the RBFSampler and SGDClassifier from the scikit-learn library [49]. We applied the four classifiers to subimages of each condition and recorded the overall accuracy and Kappa coefficients of each model in Table III. SVM has no ability to recognize any of the conditions, with the kappa coefficient for all conditions being less than 0.1. MLP exhibited the highest OA/Kappa in c2a, with values of 0.765/0.586, whereas it performed the worst in c3, with OA/Kappa values of only 0.371/0.212. RF outperformed both SVM and MLP in all conditions, with the worst performance in c1a, where OA/Kappa was 0.767/0.594. RF performed best in c3, with OA/Kappa values of 0.921/0.850, but it was still lower than the SALC metrics in this condition. OA of SALC in all conditions is above 0.89, and Kappa is above 0.8, demonstrating better classification ability compared to classical single classifiers.

Fig. 9 shows c5, corresponding manual labels, and the classification results of the four classifiers. All labeled pixels were used to calculate metrics with the classification results of SALC. All three classical classifiers are able to identify most of the leads, but each model has noticeable misclassifications in different categories. SVM incorrectly classified a large portion of the background as BL, resulting in a high recall but low precision for the BL. RF performs better than SVM in recognizing BL, but it completely misclassifies DL as MYI. Among the classifiers, MLP is the most conservative in recognizing leads and misclassifies a large number of MYI as FYI. SALC shows more accurate recognition across all categories with no significant misclassifications, and its results are most consistent with the manual labels. The classical classifiers struggle to converge perfectly when faced with the entire dataset, with RF only performing well on BL and MLP having nearly no ability to recognize MYI. In contrast, SALC is more satisfying, especially when handling large datasets, as it achieves good convergence without overfitting. This indicates that, compared to traditional single models, SALC

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TABLE II
AMOUNT OF TOTAL PIXELS FOR TRAINING AND TESTING IN EACH CONDITION (FROM THE RED BOXES IN FIG. 1)

	BL	DL	FYI	MYI
Con. 1 Train / Test	98164 / 42070	73162 / 31355	306794 / 131483	36145 / 15491
Con. 2 Train / Test	451846 / 193648	10616 / 4550	643789 / 275907	91920 / 39394
Con. 3 Train / Test	111922 / 47967	613752 / 263037	6146 / 2634	1129232 / 483957
Con. 4 Train / Test	164728 / 70598	22240 / 9532	24065 / 10314	156212 / 66948
Con. 5 Train / Test	277195 / 118798	52669 / 22572	340676 / 146004	308794 / 132340
Total set Train / test	4920068 / 2108600			

TABLE III
ACCURACY OF SALC AND TRADITIONAL SINGLE MODELS IN DIFFERENT CONDITIONS AND SUBIMAGES

	Con.1 (c1a) OA/Kappa	Con. 2 (c2a) OA/Kappa	Con. 3 (c3) OA/Kappa	Con. 4 (c4) OA/Kappa	Con.5 (c5) OA/Kappa
SVM	0.307 / 0.017	0.381 / -0.043	0.137 / -0.0001	0.298 / 0.02	0.307 / 0.021
RF	0.767 / 0.594	0.821 / 0.695	0.921 / 0.850	0.906 / 0.832	0.761 / 0.659
MLP	0.659 / 0.392	0.765 / 0.586	0.371 / 0.212	0.632 / 0.471	0.521 / 0.306
SALC	**0.893 / 0.810**	**0.947 / 0.905**	**0.932 / 0.870**	**0.937 / 0.900**	**0.907 / 0.872**

is better suited for Arctic leads recognition and solving the generalization issue of models.

C. Ablation Experiments for Model

In order to evaluate the effectiveness of each component in the SALC model, we designed an ablation study focusing on three aspects: preprocessing, feature construction, and the composition of the meta classifier.

In Section III-A, the primary factors affecting image quality are the incidence angle and thermal noise; therefore, we investigated how to handle these two factors to minimize their impact on the SALC. In order to compare the effects of each preprocessing method, we selected an image slice from image C5 (see Fig. 10). This slice is characterized by a large range variation in the azimuth direction, with roughly equal proportions of MYI and FYI, providing a clear example of variations in incidence angles. We designed six experiments where the radiometric calibration, speckle noise removal, and decibelization steps remained the same. For each experiment, a combination of different incidence angles and thermal noise processing methods was selected. The treatments for the incidence angle include incidence angle correction (IC_co), input incidence angle as a feature (IC_fe), and no treatment (IC_no), while the treatments for the thermal noise include conventional thermal noise removal (TNR_tra), which can be completed using the ESA image processing tool SNAP, advanced thermal noise removal (TNR_adv) and input as a feature (TNR_fe). For example, Exp. 1 did not treat the incidence angle and used conventional thermal noise removal, its scheme is TNR_tra + IC_no. In order to determine the optimal method for handling the incidence angle, we designed Exp. 2 through Exp. 4 for comparison. These three experiments only differ in how the incidence angle is handled. Exp. 1 and 4, as well as Exp. 3 and Exp. 5 were used to explore the optimal processing method for thermal noise. What is more, like inputting incidence angle as a feature, we designed Exp. 6.

Fig. 10 shows, from top to bottom, the slice of the subimage, the incidence angle for that region, thermal noise, and the classification results of six experimental setups. The incidence angle is consistent in the azimuth direction and varies evenly in the range direction. Thermal noise is distributed across the entire image, with more noticeable noise at the boundary of each swath and mid of the first swath (see Fig. 10). Exp. 1 performs the worst among the six experiments, and the complex condition is difficult to handle due to the effects of the incidence angle and thermal noise. The results in Fig. 10 show that only the background is roughly correct, and the leads are almost impossible to identify. Exp. 2 shows the result of the SALC model, which performs the best and is able to accurately identify almost all of the leads. Comparing Exp.1 and Exp.5,

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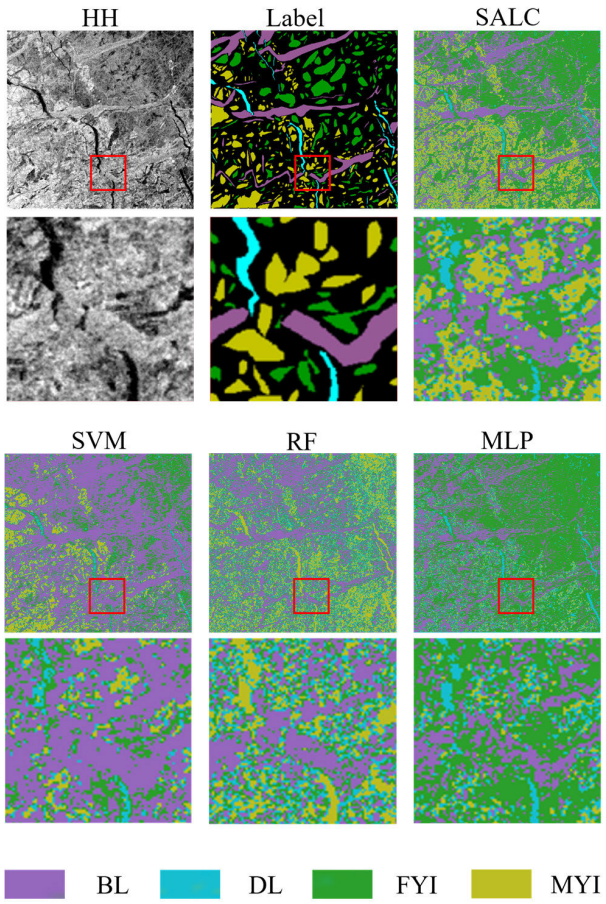


Fig. 9. Application results of SALC and traditional classifiers in subimage c5. The second rows of each group show the area within the red-boxed area.

using TNR_tra, the incidence angle correction improves the overall accuracy to 0.905 from 0.755, and most of the leads can be identified by this method. Comparing Exp.2, Exp.3, and Exp.4, using TNR_adv, the incidence angle correction has a limited effect on improving overall accuracy, while inputting the incidence angle as a feature significantly improves the overall accuracy again. As Fig. 10 shows, in Exp.3–Exp.5, the DL in region ① are misclassified as FYI, while Exp.2 correct this misclassification. This indicates that considering the impact of the incidence angle in classification is necessary, and using the incidence angle as a feature input is the optimal method for handling the incidence angle.

Comparing Exp.1 and Exp.4, both without handling the incidence angle, the advanced thermal noise removal method could also increase the overall accuracy from 0.755 to over 0.9. Comparing Exp.3 and Exp.5, with incidence angle correction applied, the advanced thermal noise removal method only increased the overall accuracy by 0.003. Neither advanced thermal noise removal nor the incidence angle correction could, however, completely eliminate the impact of thermal noise. As shown in region ② of Fig. 10, thermal noise caused the misclassification of the striped area as DL.

As shown in Table IV, when incidence angle correction was not applied, the advanced thermal noise removal method could significantly improve overall accuracy (Exp. 1 and Exp. 4). When both methods used traditional thermal noise removal, incidence angle correction could also substantially enhance accuracy (Exp. 1 and Exp. 5). Applying both methods simultaneously, however, provided only limited improvement compared to using either method individually. This suggests that both incidence angle correction and advanced thermal noise removal are beneficial for classification, but the two operations exhibit "collinearity."

In order to explore the effectiveness of each feature in the data samples, we divided the features into backscatter features (HH, HV, and ratio), texture features, and upscaled features. Different combinations of these features were used to train the model, which was applied to the image slice shown in Fig. 8. The accuracy metrics are recorded in Table V.

As shown in Table V, when only backscatter features are used as input, although the SALC's overall accuracy exceeds 0.7, the recall rate for DL is very low, indicating that relying solely on backscatter features makes it difficult to accurately identify leads. Exp. 8 and Exp. 9 add texture features and upscaled features, respectively. After adding texture features, the overall accuracy of the SALC improves significantly, and its ability to identify leads is notably enhanced. The effect of upscaled features is twofold: the recall rate for DL increases significantly, but the recall rate for BL slightly decreases, resulting in an overall better performance. Exp. 10, which is the final selected feature combination, includes both texture and upscaled features. Compared to Exp. 8, the recall for BL slightly decreases, while the recall for DL improves significantly. Exp. 7 and 9, as well as Exp. 8 and 10, demonstrate that upscaled samples have a slight effect on model improvement, whereas texture features provide a more substantial improvement. This is because upscaled samples primarily serve to augment the data, while texture features are spatial features of the pixels, providing more diverse and richer information for classification.

Finally, we designed Exp. 11–Exp. 13 to investigate the impact of different meta classifiers on the SALC. Following the standard stacking algorithm approach, we retained the strong learner MLP as the base learner to capture patterns across different conditions. In Exp.11–Exp.13, the preprocessing scheme, input features, and base learners are consistent, while the meta classifiers are SVM, RF, and XGBoost, respectively. The SVM method used here is consistent with that described in Section IV-A. The results of these three experiments are shown in Table VI. In addition, all experiments were conducted on the same computer, with a CPU of 13th Gen Intel[1] Core[2] i7-13700F 2.10 GHz and 32 GB of memory.

Among the three experiments, SVM showed the worst results. Although the precision for DL was 0.89, the recall was only 0.25, indicating that the model built with this meta classifier cannot effectively perform lead detection. Compared to RF, XGBoost showed a slight decrease in recall for BL, with the F1-score dropping by only 0.01, while other metrics showed significant improvement. Additionally, XGBoost had the fastest training time at only 66 s, which is 50 times faster

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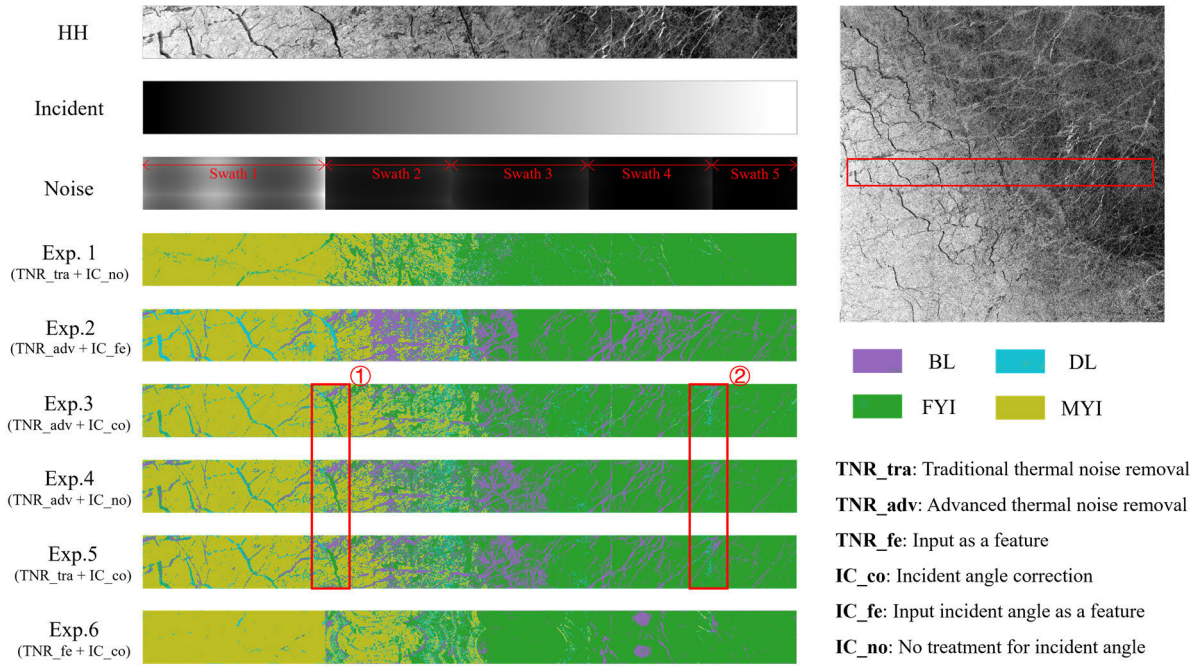


Fig. 10. Classification results of the slice in image C5 with different preprocessing methods. The incidence angle and thermal noise processing modules are IC_no: incidence angle correction, IC_fe: input incidence angle as a feature, IC_no: no treatment to incidence angle, TNR_tra: traditional thermal noise removal, TNR_adv: advanced thermal noise removal, and TNR_fe: input thermal noise as a feature.

TABLE IV
IMPACT OF DIFFERENT PREPROCESSING METHOD ON SALC

	scheme	Precision (BL/DL)	Recall (BL/DL)	F1-score (BL/DL)	Kappa	OA
Exp.1	TNR_tra + IC_no	0.48/0.81	0.12/0.01	0.19/0.02	0.6028	0.755
Exp.2	**TNR_adv + IC_fe**	**0.94/0.96**	**0.90/0.97**	**0.92/0.96**	**0.9603**	**0.974**
Exp.3	TNR_adv + IC_co	0.81/0.70	0.79/0.54	0.81/0.62	0.8617	0.909
Exp.4	TNR_adv + IC_no	0.78/0.66	0.76/0.46	0.77/0.54	0.8501	0.901
Exp.5	TNR_tra + IC_co	0.81/0.70	0.79/0.52	0.80/0.59	0.8564	0.905
Exp.6	TNR_fe + IC_co	0.72/0.79	0.16/0.11	0.26/0.19	0.6641	0.799

than RF, indicating that XGBoost is the most suitable meta classifier among these three classifiers.

V. DISCUSSION

A. Sensitivity Assessment of SALC to Erroneous Samples

All the labels used for training were manually marked based on visual interpretation. Although we enriched SAR image interpretation knowledge and understood the characteristics of leads at different developmental stages in SAR images before labeling, it was still difficult to avoid errors, especially for ambiguous sea ice/lead areas; therefore, we designed this experiment to explore the model's sensitivity to erroneous samples, specifically its tolerance for incorrect labels. We assumed that the error rate of the training samples was 0 and intentionally modified a portion of the samples to introduce errors, with error rates set at 10%, 20%, 30%, 40%, and 60%. In order to make erroneous samples, we randomly selected a certain percentage of the data, altered their labels to wrong categories, and returned them to the original dataset for training. As shown in Table VII, when the error rate was 0, the model's overall accuracy was 0.933, and the Kappa coefficient was 0.894. As the error rate increased, the model's performance declined.

When the error rate reached 20%, accuracy dropped to 0.754, and the Kappa coefficient decreased to 0.600, still indicating that the model fits well. When the error rate reached 40%, the model, however, performed poorly. Increasing the error rate to 60%, the Kappa coefficient dropped below 0.2, and the model was considered to have failed to converge. The model's performance remained excellent when the error rate was below 15%, and we are confident that the manual labeling error rate cannot exceed 15%. We, therefore, conclude that

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TABLE V
IMPACT OF DIFFERENT FEATURES ON SALC

	Feature	Precision (BL/DL)	Recall (BL/DL)	F1-score (BL/DL)	Kappa	OA
Exp.7	Backscatter	0.71/0.71	0.63/0.37	0.67/0.49	0.6009	0.727
Exp.8	Backscatter + Texture	0.86/0.91	0.85/0.81	0.86/0.86	0.8311	0.883
Exp.9	Backscatter + Upscaling	0.70/0.58	0.56/0.60	0.62/0.59	0.6426	0.739
Exp.10	**Backscatter + Texture + Upscaling**	**0.86/0.91**	**0.83/0.90**	**0.84/0.91**	**0.8720**	**0.907**

TABLE VI
IMPACT OF DIFFERENT META CLASSIFIERS ON SALC

	Meta classifier	Precision (BL/DL)	Recall (BL/DL)	F1-score (BL/DL)	Kappa	OA	Time(s)
Exp.11	SVM	0.61/0.89	0.66/0.25	0.63/0.39	0.5401	0.693	106.33
Exp.12	RF	0.81/0.90	0.90/0.83	0.85/0.87	0.8203	0.875	3282.72
Exp.13	**XGBoost**	**0.86/0.91**	**0.83/0.90**	**0.84/0.91**	**0.8720**	**0.907**	**66.13**

TABLE VII
SENSITIVITY ASSESSMENT OF SALC TO ERRONEOUS SAMPLES

Error rate	0	10%	**15%**	20%	30%	40%	60%
OA	0.933	0.833	**0.803**	0.754	0.654	0.564	0.385
Kappa	0.894	0.740	**0.668**	0.600	0.470	0.351	0.140

SALC is capable of handling a limited amount of errors in the labeling process.

B. Limitations of SALC

This study proposes a SALC method aiming to identify leads in Sentinel-1 SAR images of different Arctic conditions. We have verified the effectiveness of this method, but there are still some limitations in accurately identifying leads across the entire Arctic.

When the backscatter and spatial texture of pixels are similar, the classifier may struggle to make accurate classifications. This situation often occurs due to the confusion between FYI and DL, as well as between FYI and BL. Fig. 11 illustrates an example where DL is easily confused with FYI. In Fig. 11(b) and (d), the same DL area and FYI area are annotated. As shown in Fig. 11(d), the FYI has a smooth ice surface, which results in low backscatter values in Fig. 11(b). In the DL area, the backscatter values are statistically as follows: HH polarization ranges from −24.9 to −17.2 dB, with an average of −21.3 dB; HV polarization ranges from −33.9 to −28.6 dB, with an average of −30.7 dB. In the FYI area, the HH polarization ranges from −24.5 to −17.8 dB, with an average of −20.7 dB, and the HV polarization ranges from −34.3 to −27.6 dB, with an average of −30.6 dB.

Similarly, Fig. 12 demonstrates an example where BL and FYI are easily confused. From Fig. 12(b), it is visually apparent that the BL area and FYI area exhibit very similar characteristics. Statistically, in Fig. 12, the HH polarization in the BL area ranges from −22.1 to −17.4 dB, with an average of −19.4 dB, while the HV polarization ranges from −39.3 to −28.1 dB, with an average of −34.9 dB. In the FYI area of Fig. 12, the HH polarization ranges from −25.1 to −15.6 dB, with an average of −20.7 dB, and the HV polarization ranges from −42.3 to −25.5 dB, with an average of −34.1 dB.

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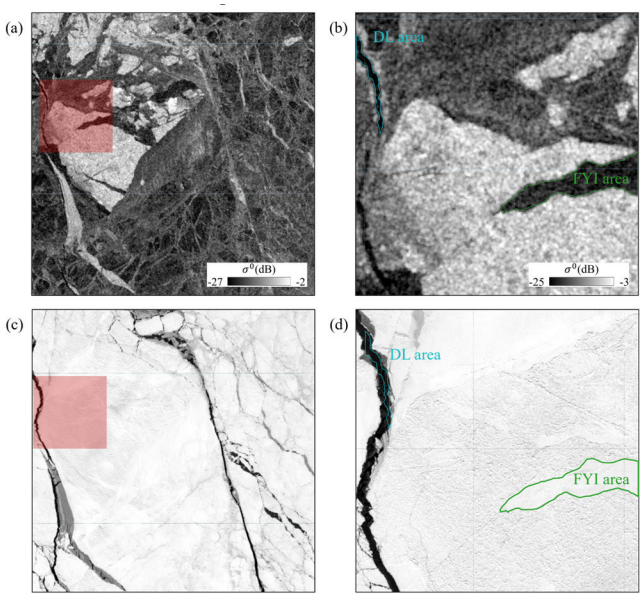


Fig. 11. Instance where DL and FYI are easily confused in the application of SALC. (a) and (c) HH image and Sentinel-2 image, respectively, of the same region. (b) and (d) Correspond to the red masked regions in (a) and (c), respectively.

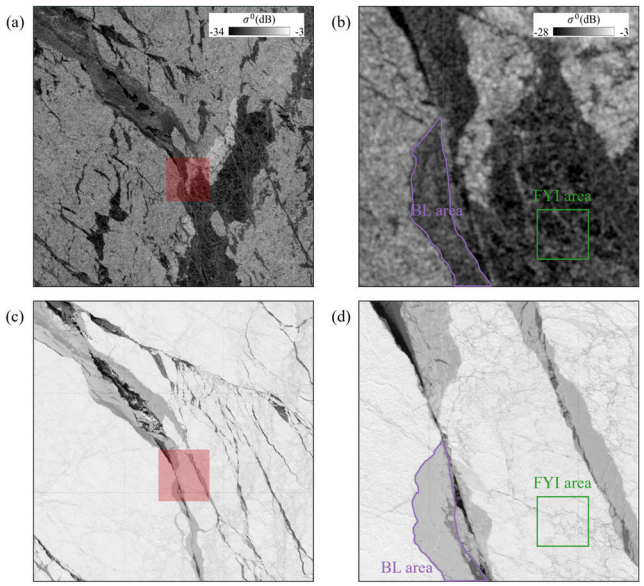


Fig. 12. Instance where BL and FYI are easily confused in application of SALC. (a) and (c) HH image and Sentinel-2 image, respectively, of the same region. (b) and (d) Correspond to the red masked regions in (a) and (c), respectively.

In both examples, the backscatter values of the leads, whether in HH or HV polarization, are extremely similar to those of FYI, making it difficult to distinguish them without the aid of optical imagery. This confusion arises primarily from two factors. First, the developmental stages of sea ice are continuous, making it challenging to directly determine the specific stage from the imagery. Second, different ice cover types may exhibit similar features in SAR images, akin to the "same object with different spectra" phenomenon observed in optical imagery.

Additionally, SALC is a pixel-based classification method that integrates backscatter features and spatial information, which enables it to detect most small leads. Pixel-based classification methods can, however, produce more fragmented patches in the classification results due to the speckle noise in SAR images. Postprocessing of the classification results is, therefore, necessary when applying this method.

VI. Conclusion

In this article, we proposed a SALC to automatically detect Arctic winter leads via Sentinel-1 imagery. We classified Arctic sea ice backgrounds based on MYI frequency and divide leads into BL and DL according to their development stages. Lead conditions were divided to five categories, and effective data preprocessing methods were used to correct for the influence of incidence angle and thermal noise. The proposed SALC method uses a two-layer structure with MLP as the base learner and XGBoost as the meta classifier, demonstrating strong performance in lead detection under various conditions.

Compared to classical classifiers, SALC outperforms SVM, RF, and MLP in accuracy and robustness, especially in complex conditions like C5, where FYI and MYI are mixed, and the stages of leads development are diverse. In this condition, the overall accuracy/Kappa of SALC is 0.907/0.872, whereas RF, which performed best among the three classical single classifiers, achieved only 0.761/0.659. Through ablation experiments, we identified the optimal preprocessing scheme: inputting the incidence angle as a feature and correcting thermal noise by an improved removal method. Compared to the traditional thermal noise removal and no treatment for the incidence angle, the overall accuracy increased from 0.755 to 0.974. Additionally, using the incidence angle as input parameter yields clearly better results than correcting for its influence. We have also verified that texture and upscaling features, and XGBoost also contribute to the accuracy of SALC. Given that SALC performed well at a 15% sample error rate, it has shown its ability to handle errors that occur in sample labeling; however, SALC still has several limitations. It may misclassify sea ice and leads with similar backscatter and texture and mistakenly identify ice ridges as leads. In summary, SALC is an effective and reliable method for lead detection tasks, with the potential for Pan-Arctic lead mapping using Sentinel-1 SAR image.

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